

TWO KEY METHODS, EXPLAINED

A closer look under the hood

1

Fixing the churn label

Why the data's "Churn probability?" column was unusable — and how we built a trustworthy one.

2

Choosing 4 groups

What the "silhouette" score measures, and why the tidiest answer wasn't the one we picked.

First, what is a “label” — and why did ours need fixing?

In plain terms

A “label” is the answer key the model learns from.

To teach a model to spot leavers, we must first show it thousands of past riders each marked “stayed” or “left.” That marking is the label. If the answer key is wrong, everything the model learns from it is wrong too — like revising from a textbook full of errors.

What we were given

The data arrived with a ready-made “chance of leaving” number for each rider. Convenient — so we tested it.

We compared riders it marked as leaving against those it marked as staying. They looked **exactly the same**: same ride frequency, same time since last trip, same spend. The number was disconnected from real behaviour.

The tell-tale test

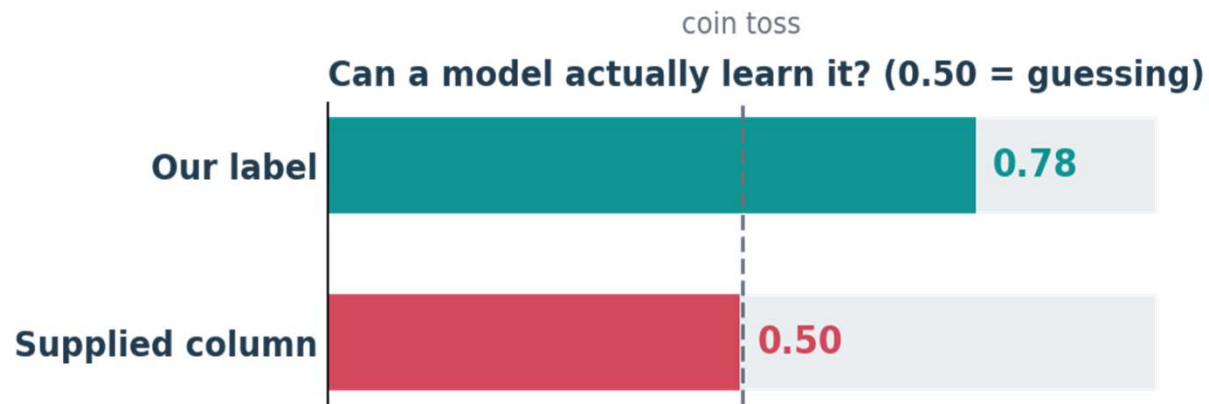
Days since last ride — normally the strongest warning sign anywhere:

Marked “stayed”: 18 days

Marked “left”: 18 days

Basically identical. A genuine leaver should have been away far longer. This column simply didn’t reflect who was leaving.

A model given the old label learns nothing



0.50 means pure guessing.

We scored how well a model could learn each label, from 0.50 (a coin toss) to 1.00 (perfect).

The supplied column scored **0.50** — literally no better than flipping a coin. There was simply no pattern in it to find.

This is the moment we decided the label had to be rebuilt from real behaviour, not accepted as given.

How we built a label we could trust

We defined “leaving” by what riders actually did next.

Simple and honest: we froze time on one day, then looked at the following month. If a rider took no trip in that month, we marked them as having left. Real behaviour, not a mystery number.

Then we made the label realistic, on purpose

Because the data is synthetic, we gently shaped who leaves so it follows real-world common sense — the same rules any rider-retention expert would expect:



Gone quiet lately (Recency)

more likely to leave



Rides often (Frequency)

more likely to stay



Spends more (Monetary Value)

more likely to stay



Hit by surge prices

more likely to leave



Higher loyalty tier

more likely to stay



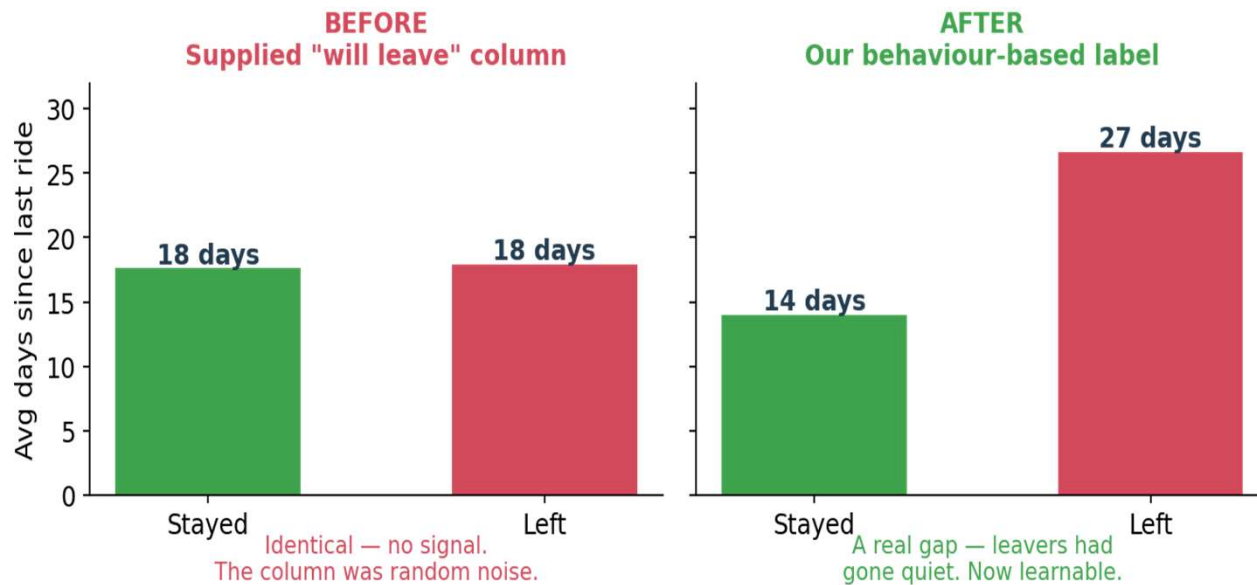
Gives low ratings

more likely to leave

We also added a pinch of randomness — because in real life, behaviour only ever hints at what someone will do; it never perfectly predicts it.

PART 1 · THE RESULT

Now “stayed” and “left” riders actually look different



A real, sensible gap appears.

Leavers had gone quiet — 27 days since a ride versus 14 for those who stayed. Exactly the kind of difference the team would expect to see.

And now a model can learn it.

The learnability score jumped from 0.50 (coin toss) to 0.78 — a strong, realistic signal.

Honest by design: we could have made it score higher — but a suspiciously perfect result would mean we'd faked the problem away.

“Isn’t shaping the label just cheating?”

Q

Did you tell the model the answer?

No. The label is built only from things known before our freeze-date. The model never sees the future it’s trying to predict.

Q

Did you make it look too good?

No. We deliberately kept it realistic (0.78, not 0.99). A near-perfect score would be the red flag.

Q

Can anyone check your work?

Yes. One setting rebuilds the original, unusable label and reproduces the 0.50 result — so the whole thing is transparent and reversible.

Q

Why was this needed at all?

Only because the sample data is synthetic and shipped without real leaving-patterns. On live data, the next-month definition alone would carry genuine signal.

What does “grouping” even mean here?

We sort 10,000 riders into a few look-alike groups.

Instead of treating every rider individually, we bundle together riders who behave similarly — by how recently they rode, how often, and how much they spend (**RFM**). Marketing can then design a handful of strategies instead of ten thousand. The computer finds the groupings; we decide how many.

The one puzzle: how many groups?

Too few and everyone’s lumped together uselessly; too many and you can’t act on them all. We need a way to judge how “tidy” a given number of groups is. That measure is called the silhouette score.

In plain terms

Silhouette = how well each rider fits their own group.

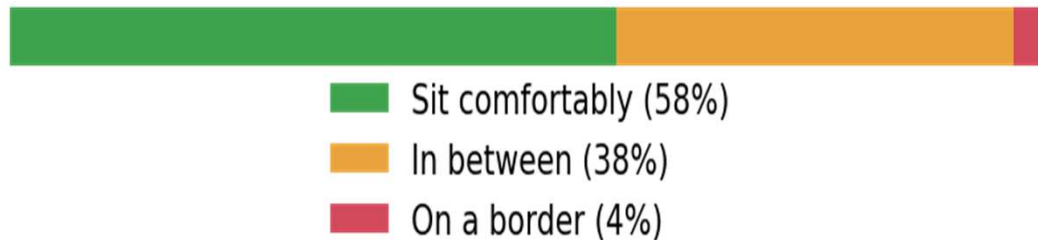
For every rider it asks two things: how close are they to their own group, and how far from the nearest other group? If they’re snug in their own and far from the rest, they “fit” well. Average that over everyone and you get one score from roughly 0 (messy, overlapping groups) to 1 (crisp, well-separated groups).

Think of tables at a party

A picture for it

Imagine guests seated at tables. A guest is **well-placed** if they have loads in common with their own table and little with the next table over. A guest who'd fit just as well at the table behind them is **on a border**. The silhouette score is just the average “how well-placed is everyone?” across the whole room.

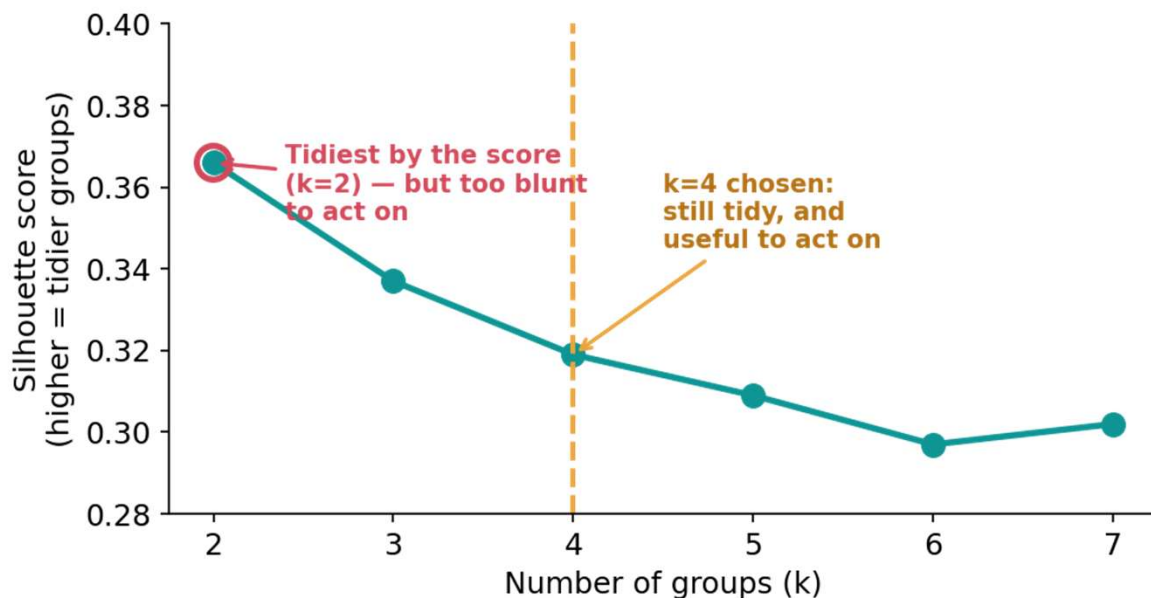
With 4 groups: do riders fit their group?



Our 4 groups are tidy.

With four groups, 58% of riders sit comfortably in theirs and only 4% are on a border. That's a clean, dependable split — the groups are real, not forced.

The tidiest answer wasn't the useful one



2 groups scored highest. We chose 4.

The score technically peaked at two groups. But “everyone is either high- or low-value” is too blunt to build campaigns on. **Four groups score almost as tidy while giving genuinely different, actionable audiences.**

The score advises; people decide.

A tidiness score can't know what's useful to the business. That judgement stays with the analyst — and it's worth making consciously.

PART 2 · THE RESULT

Four groups that each need a different response



Our best riders

26% of riders

Loyal, high-spending, rarely leave. Protect — don't discount.



Newcomers

21% of riders

Recently joined, still forming a habit. Encourage, don't bribe.



Starting to fade

35% of riders

Slipping in activity. Watch closely; nudge the fading ones.



About to leave

19% of riders

Most of our churn sits here. This is where the budget goes first.

Same idea as Part 1: the method is only worth it because it changes what we do — here, four audiences instead of one blanket message.

THE TWO TAKEAWAYS

1 · Fixing the churn label

The supplied “will they leave?” column was random noise (scored 0.50 — a coin toss). We rebuilt it honestly from real behaviour — no trip next month = left — and shaped it with common-sense rules. Result: a realistic, learnable **0.78**, and fully reproducible.

2 · Choosing 4 groups

The silhouette score rates how tidily riders fit their groups (0 messy → 1 crisp). It peaked at 2 groups, but 2 is too blunt to act on. We chose 4 — nearly as tidy (**58% sit comfortably**) and genuinely useful. The score advises; the analyst decides.

Both come down to one principle: be honest about the method, and judge it by whether it helps us act.